

Flat equiangular tight frames are explicit quasi-symmetric designs

Autonomous proof packet for arXiv:1602.05557

11 August 2026

Target and result

Problem 2 of Fickus–Mixon–Jasper [1] asks, for even q , whether there are $q(q^2 + q - 1)$ vectors with entries in $\{\pm 1\}$ that form an ETF for $\mathbf{1}^\perp \subset \mathbb{R}^{q(q+1)}$ and have pairwise inner products of absolute value q . Their first open case is $q = 6$.

in light of Theorem 3, Problem 1 is equivalent to:

Problem 2. Given q even, does there exist a sequence $\{\varphi_i\}_{i=1}^{q(q^2+q-1)}$ of $\{\pm 1\}$ -valued vectors that form an ETF for $\mathbf{1}^\perp$ in $\mathbb{R}^{q(q+1)}$, and so have $|\langle \varphi_i, \varphi_j \rangle| = q$ for all $i \neq j$?

The answer to Problem 2 is “yes” when $q = 2$, e.g. (4), and “no” when $q = 4$, since real ETFs with $(d, n) = (19, 76)$ do not exist [1], [33]. To our knowledge, the $q = 6$ case is open. Here, $(m, n) = (42, 246)$ and so computer-assisted searches might be impractical. Cases like this where $\frac{n}{m} = q \equiv 2 \pmod{4}$ are particularly interesting since they might be related to a fact from [16]: if there exist $m \times n$ real flat matrices whose columns

We give a constructive equivalence with one sharply specified object from design theory.

Theorem 1. *Let q be even and set*

$$m = q(q+1), \quad v = m - 1 = q^2 + q - 1, \quad b = qv.$$

The following are equivalent.

1. *There is a matrix $\Phi \in \{\pm 1\}^{m \times b}$ whose columns form an ETF for $\mathbf{1}^\perp \subset \mathbb{R}^m$ and whose distinct column inner products have absolute value q .*
2. *There is a quasi-symmetric design on v points with b blocks and parameters*

$$k = \frac{q(q+1)}{2}, \quad r = \frac{q^2(q+1)}{2}, \quad \lambda = \frac{q^2(q+1)}{4},$$

whose two block-intersection numbers are

$$x_1 = \frac{q^2}{4}, \quad x_2 = \frac{q(q+2)}{4}.$$

Both transformations are explicit and inverse up to row/column switching.

For $q = 6$, Problem 2 is therefore equivalent to a quasi-symmetric 2-(41, 21, 63) design with 246 blocks, replication 126, and block intersections 9 and 12.

Proof: ETF to design

Assume (1). Since the b columns have squared norm m and form a tight frame for the $v = m - 1$ dimensional space $\mathbf{1}^\perp$, their frame operator is

$$\Phi\Phi^\top = \frac{bm}{v} \left(I - \frac{1}{m}J \right) = q^2(q+1)I - qJ. \quad (1)$$

Multiply each column by its first entry. This makes the first row all +1 and preserves every hypothesis. For a column j , let B_j be the set of indices among the remaining v rows where its entry is -1 .

Every column is orthogonal to $\mathbf{1}$, so

$$|B_j| = \frac{m}{2} = k. \quad (2)$$

Every other row has inner product $-q$ with the first row by (1), hence has

$$r = \frac{b+q}{2} = \frac{q^2(q+1)}{2} \quad (3)$$

negative entries. Any two such rows also have inner product $-q$; their negative supports therefore have symmetric difference $(b+q)/2 = r$ and intersection

$$\lambda = r - \frac{r}{2} = \frac{q^2(q+1)}{4}. \quad (4)$$

Thus the B_j form a $2-(v, k, \lambda)$ design with replication r .

If $t = |B_i \cap B_j|$, the corresponding column inner product is

$$1 + v - 4(k - t) = 4t - m.$$

Its absolute value is q , so $t = (m - q)/4 = q^2/4$ or $t = (m + q)/4 = q(q + 2)/4$. This proves (2).

Proof: design to ETF

Conversely, let N be the $v \times b$ incidence matrix of the design and set

$$\Phi = \begin{bmatrix} \mathbf{1}^\top \\ J - 2N \end{bmatrix}.$$

Equation (2) makes every column sum equal to $m - 2k = 0$. Two columns whose blocks meet in t points have inner product $4t - m$, hence $\pm q$.

The first row has inner product $b - 2r = -q$ with every other row. Two distinct rows of N meet in λ blocks, so their signed versions have inner product

$$b - 4(r - \lambda) = -q.$$

All row norms are b , and consequently

$$\Phi\Phi^\top = (b+q)I - qJ = q^2(q+1) \left(I - \frac{1}{m}J \right). \quad (5)$$

Thus Φ has rank $m - 1$ and is tight precisely on $\mathbf{1}^\perp$. Its column coherence is $q/m = 1/(q+1)$, while

$$\frac{b - (m - 1)}{(m - 1)(b - 1)} = \frac{1}{(q + 1)^2};$$

so this is the Welch value. The columns are an ETF, proving (1). $\square$

Block graph and the first open case

Let A join two blocks when they meet in x_2 points. From

$$N^T N = (k - x_1)I + (x_2 - x_1)A + x_1 J, \quad N N^T = (r - \lambda)I + \lambda J,$$

one obtains a strongly regular block graph with degree

$$K = \frac{(q-1)(q+1)(q+2)}{2},$$

restricted eigenvalues $(q^2 - 2)/2$ and $-(q + 2)/2$, and parameters

$$\Lambda = \frac{(q+4)(q^2-2)}{4}, \quad M = \frac{q^2(q+2)}{4}.$$

For $q = 6$ this is $\text{SRG}(246, 140, 85, 72)$ with restricted eigenvalues 17, -4 . The standard integrality, absolute, complement, and Krein conditions all pass. Moreover, recent general restrictions on quasi-symmetric designs [3] depend here on invariants of the actual unknown graph. The equivalence therefore sharpens the finite target but does not settle its existence.

Proof intuition

The extra flatness in Problem 2 is decisive. After one harmless column switch, every vector is the signed incidence vector of a half-size block. ETF tightness controls row intersections, producing a 2-design; ETF equiangularity controls column intersections, making that design quasi-symmetric. Conversely those two intersection laws reproduce the entire row and column Gram data of the ETF. The hard $q = 6$ question is now a single finite incidence problem rather than a continuous frame search.

References

- [1] M. Fickus, D. G. Mixon, J. Jasper, *Equiangular tight frames from hyperovals*, arXiv:1602.05557v2.
- [2] A. Barg, K. A. Okoudjou, W.-H. Yu, *Finite two-distance tight frames*, arXiv:1402.3521v2.
- [3] B. Bagchi, *Parametric restrictions on quasi-symmetric designs*, arXiv:2102.05576v2.